How Can Filecoin's Bandwidth Measurement System be Optimized?

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1. INTRODUCTION

In decentralized storage networks, trust must often be established without direct oversight. Unlike large cloud storage providers such as Amazon or Google, which operate their own data centers directly, decentralized systems rely on a globally distributed set of independent storage providers to store and retrieve data. For these systems to be competitive against large cloud providers, the independent storage providers must be capable of handling consistent bandwidth demand and supply reliable retrieval speeds.

Filecoin, a blockchain-based decentralized peer-to-peer storage network, avoids this reliance on trust by implementing a bandwidth measurement system (BMS). The BMS is designed to evaluate the available bandwidth of a storage node. Filecoin storage nodes can participate in retrieval contracts, where they return requested files to a client. Clients pay these storage nodes to store and retrieve files, and BMS is essential to guarantee that a client is receiving the performance they purchased. To preserve fairness and prevent gaming of the system, these measurements are conducted adversarially, without the storage provider's awareness. This adversarial strategy ensures that reported performance metrics reflect realistic operating conditions rather than optimized responses to known tests.

This paper explores a way in which the BMS can reduce the cost of measurement while preserving the accuracy of its reported measurements. First, the modifications to the BMS must maintain or improve the accuracy of the reported bandwidth measurements. Next, this system must also be efficient with the volume of data and duration of time it uses for its measurements. The time of measurement sessions is especially important as this system attempts to saturate the provider's available bandwidth. Unlike in isolated testing environments, Filecoin's measurement system operates alongside real traffic. As a result, bandwidth consumed by measurement workers may temporarily reduce the available capacity for genuine file retrievals, affecting end-user experience. Minimizing the duration of time a measurement takes is essential to preventing unnecessary congestion for actual end-users.

Also, with more time of measurement the monetary cost of each measurement increases. The bandwidth measurement system uses Amazon's Elastic Container Service to gather its measurements which, similar to other container services, charges based on the amount of time that the containers are used. We found that our technique results in reported bandwidths **within a factor of 2 89.5%** of the time and an average of a **16% runtime reduction** per measurement**.** If this time could be reduced, the average cost per measurement could be reduced. This paper investigates whether the DHP Early Exit strategy optimizes Filecoin's bandwidth measurement system by producing more accurate and cost efficient bandwidth measurements.

2. DATASETS AND DATA PROCESSING

The Filecoin bandwidth measurement system, or BMS, is Filecoin's method of determining the available bandwidth of a storage provider on the Filecoin network. As outlined earlier, it is essential that the BMS measures without the storage provider's knowledge.

In most internet speed tests, the host machine which wants to be tested requests a file from a storage provider (like Cloudflare). As this file is being downloaded by the host machine, the host measures their own throughput. This method assumes that the host machine is the bottleneck in the download. The intent of Filecoin's bandwidth measurement system is to ensure that the bottleneck in the test is the storage node, not the measurement system's host machine that initiated the download.

To effectively avoid the BMS from measuring its own available bandwidth (to prevent itself from becoming the bottleneck of the measurement), it uses virtual machines (from AWS Elastic Container Service) to target a storage node. For a majority of the measurements recorded, 10 virtual machines, known as *workers*, are used. These workers simultaneously download the same file from the bandwidth measurement system simulating multiple users requesting the same file. Figure 1 is a simplified diagram of the method described above.

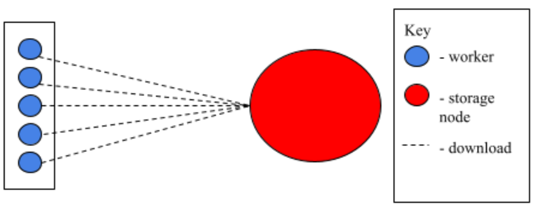

Figure 1: Simplified structure of a measurement session in BMS

Figure 1 depicts a single measurement of a targeted storage provider. Up to three (3) of the Figure 1 measurement sessions occur in succession in a *job*. The recorded throughput of each worker is then aggregated by sum to calculate the theoretical bandwidth of the storage node. The job data is gathered through the bandwidth measurement system's API.

3. FAULTS IN THE CURRENT METHOD

For each *measurement session*, BMS calculates the download speed which attempts to represent the maximum available bandwidth of the storage node. The current method of bandwidth calculation in the bandwidth measurement system is represented by the following equation:

$$\sum_{\#\, workers}^{1} (total\ bytes)/(elapsed\ seconds\ \times\ 1024^{2})$$

A question arises from this method of calculation, *why not use the maximum reported bandwidth?* Often in production networks TCP/IP is the primary congestion control used. When multiple TCP connections are targeting a single source, "it is likely that one connection will take more bandwidth due to errors in the base station" (Zhou H. et. al.). These errors can result in the TCP/IP throughputs of separate workers peaking at separate times during the simultaneous download. This will increase the reported bandwidth to an unrepresentative value.

Instead of the maximum value, the average value may provide a representative picture of a worker's performance throughout its download. It attempts to reduce noise in the measurement. This method, although efficient and relatively accurate, falls short in two key areas.

3.1 Bandwidth inflation

The current method of calculation that BMS uses does not consider the variation in end times of coordinated *workers* within a *measurement session*. For many downloads, once the first worker completes its download of 100MB, the remaining workers reclaim the available bandwidth. This reclaiming of bandwidth occasionally leads to an artificial inflation of the measurement. The following diagram illustrates this issue.

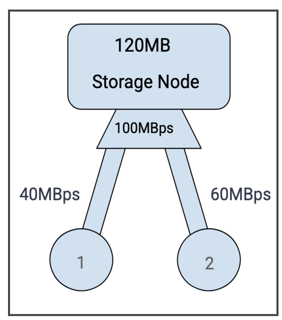

Figure 2: Example of faults in the BMS current measurement calculations. Two workers retrieving a 120MB file from a single 100MBps storage node. On the sides of each worker's pipe is their available bandwidth.

In the simplified scenario depicted by Figure 2, Worker 2 completes its download in two (2) seconds. In that same period of time Worker 1 downloaded 80MB of the 120MB file. Since Worker 1 reclaimed the available bandwidth from Worker 2, it completed the last 40MB of its download in 0.4 seconds. Based on the BMS method of calculating bandwidth, Worker 1 reports a download speed of 50MBps and Worker 2 reports a download speed of 60MBps. This results in the storage node's total available bandwidth appearing as 110MBps.

3.2 Unnecessary testing length

Although in some cases bandwidth inflation may occur, it may not always occur. Occasionally, the remaining workers do not reclaim the available bandwidth. The reported bandwidth value in these cases does not inflate, and may even deflate. These instances should still be considered unrepresentative of the bandwidth measurement as only a subset of the workers are measuring (implying that the available bandwidth may have changed due to outside interference). If these tests can be considered unrepresentative, then it should be possible to shorten the testing length by exiting a measurement session at the first instance of a worker's download completing.

4. DHP EARLY EXIT

DHP Early Exit (EE) refers to a method where the measurement session ends once the first worker completes their download. This method kills the current worker processes with the goal of maintaining, and occasionally improving the relative accuracy of the reported measurements and simultaneously reducing the time and volume per measurement for middle to high bandwidth nodes.

Storage nodes in the Filecoin network can be classified by their available bandwidth. Low bandwidth nodes report speeds of 1 Mbps to 100 Mbps, medium bandwidth nodes report between 100 Mbps and 500 Mbps, high bandwidth nodes have an available bandwidth speed of 500 Mbps to 1 Gbps, and very high bandwidth nodes have an available bandwidth from 1 Gbps to 3 Gbps.

For this study, the values used to determine the time to exit are the second by second logged values which each worker stores during its download. EE is simulated by ignoring all reported values after the one worker within a measurement session completes its download. On this simulated data, the method of calculating the saturated bandwidth value (from bytes to Mbps) is the same method used by BMS.

The EE method can be used after measurements are collected by BMS or can be directly integrated into the measurement pipeline by increasing the coordination among the workers. It is important to note that the following graphics and values are estimations and are not absolutely representative of the worker's behavior. These simulations are created using the second by second instantaneous throughputs reported by the workers as they are downloading their files. When considering the download of a file across the internet a second is a long time, therefore; the values determined by these simulations will have a margin of error.

4.1 Methodology

To evaluate the accuracy and cost-effectiveness of DHP Early Exit, we compare the reported bandwidth and elapsed download time against Filecoin's original BMS approach. Accuracy is defined by the difference in the reported bandwidth before and after the first download is completed. Cost-effectiveness is evaluated in terms of average download time per measurement session.

4.2 Simulation Details

To determine the behavior of a measurement across time, we parsed the second by second logs from 329 measurement sessions. Each worker reports a second by second log, with every log containing a timestamp, the interval bytes downloaded from the previous log, and the total bytes downloaded from the beginning of the download. For each simulation, we identified the earliest completion time across all workers and truncated the download logs of all workers at that time. The truncated logs were then used to recalculate total bytes transferred and derive the simulated bandwidth using the original BMS formula.

Of these 329 recorded measurement sessions, 131 were excluded. 125 excluded tests are instances where no workers completed the 100 MB download, and therefore DHP Early Exit in its current form can not be simulated. The remaining 9 excluded tests are due to errors within the BMS coordination between workers resulting in measurement session durations exceeding 63 seconds, 3 seconds longer than the expected hard cutoff.

4.3 Results

This section reports the outcomes of simulating DHP Early Exit across 198 measurement sessions. All metrics were calculated by using the second-by-second-logs, truncated at the earliest worker completion time per simulation. The performance of DHP Early Exit is compared to the original BMS approach in two ways: bandwidth accuracy and measurement duration.

We found that the average time saved per measurement session was 5.78 seconds, with the average measurement taking 36.14 seconds without early exit. The total time of all measurements was 180 minutes and 22 seconds, and this time reduction resulted in a total of 28 minutes and 58 seconds saved across all simulations.

4.3.1 Bandwidth Accuracy Comparison

Across all simulated measurement sessions, the bandwidth under DHP Early Exit was, on average, reported **10.28%** lower than the original BMS method. To better represent this cumulative difference, the storage nodes are broken into their respective classifications.

Node type	Average Reduction ps)
Low (1 Mbps - 100 Mbps)	0.59
Medium (100 Mbps - 500 Mbps)	29.41
High (500 Mbps - 1 Gbps)	58.07
Very High (1 Gbps - 3 Gbps)	522.45

Table 1: Node types and their average reduction in bandwidth (Original BMS Method – DHP Early Exit)

From Table 1, it is clear that the relative difference in bandwidth between the current method and DHP Early Exit is minimal. This reduction reflects the exclusion of inflated throughput caused by workers reclaiming available bandwidth after the earliest download completes. Figure 3 illustrates the behavior of one representative measurement.

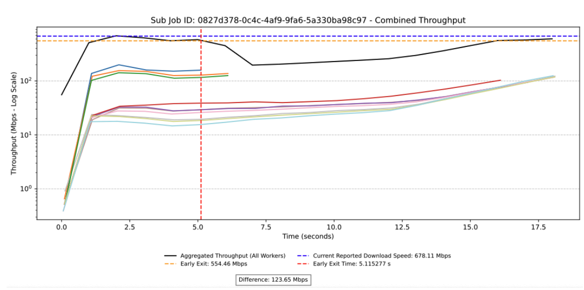

Figure 3: Instantaneous throughput over time per worker, with comparison of total bandwidth under standard BMS and Early Exit thresholds

The y-axis of Figure 3 is the instantaneous bytes per second recorded by the second by second logs on a logarithmic scale. The x-axis is time in seconds from the beginning of the download.

Each colored solid line represents an individual worker's second by second instantaneous throughput. The solid black line is the aggregated sum instantaneous throughput of all workers within the measurement session. The vertical dashed red line is the cutoff time for the DHP Early Exit calculations, the time when one worker finishes its download. The horizontal dashed blue line is the reported bandwidth calculated by the bandwidth measurement system, and the horizontal dashed orange line represents the bandwidth calculated by using DHP Early Exit.

In Figure 3, the Early Exit occurred at 5.1 seconds into the download. After this point, the remaining workers reclaimed the available bandwidth resulting in a reported bandwidth 123.65 Mbps higher than the simulated DHP Early Exit. The simulated DHP Early Exit download terminated in 5.1 seconds rather than the 18 seconds of the current method. In addition to time, DHP EE also reduced the total volume of data downloaded by 73.5 MB.

Figure 4 presents a scatter plot of reported bandwidth across all 329 evaluated measurement sessions. In general, the difference between the two methods increases with the measured bandwidth. Green points denote measurements in which no worker completed the download, and thus DHP Early Exit could not be appealed. The maximum reported bandwidth among incomplete downloads was 109 Mbps.

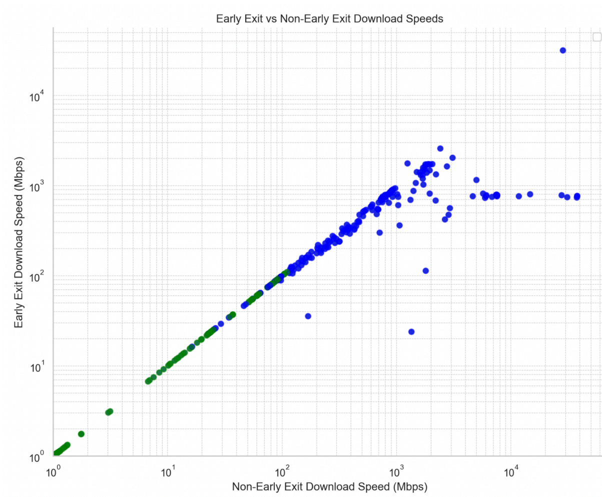

Figure 4: Scatter Plot comparison between DHP Early Exit and Non-Early Exit Download Speeds in Mbps

These findings led to a corollary question of determining which file sizes are optimal for recording an accurate download speed. The intention is to simultaneously minimize the amount of computational power in use per measurement to optimize costs.

In Figure 4, it appears that as the bandwidth increases, the discrepancy between the current and new methods increases. It also appears to be an unpredictable discrepancy. The values in the 10^4 range (X-axis) are AWS storage nodes (us-east) measured by workers in us_east (19 points ranging from *6 Gbps* to *37 Gbps*). The variability in this measurement is due to workers completing a download before others have started. DHP Early Exit does not solve the very high bandwidth inflation problem. These measurements are described in the BMS repository README as a controlled base case to ensure that the measurement system is behaving correctly, and are not measuring the output bandwidth from the AWS data center.

4.3.2 Download Time and Volume Reductions

In addition to lower bandwidth measurements, DHP Early Exit also resulted in reduced measurement time. Across all simulated measurement sessions, **47.8%** terminated their measurements in less time when compared to the current method employed by the BMS. The accumulated time save totaled **28 minutes and 58 seconds** compared to the original method with an accumulated time of **180 minutes and 22 seconds**.

Figure 5 compares elapsed time of measurement session under both methods for each. The x-axis represents the time of measurement under the standard method, and the y-axis under DHP Early Exit. Points on the diagonal line indicate equal duration. A visible concentration of points below the diagonal confirms reduced duration in many cases. A vertical cutoff at roughly 60 seconds on the x-axis which reflects a timeout enforced by the measurement system. Downloads which exceeded a time of 63 seconds due to errors in the BMS worker coordination are excluded from this graph as well. Often, these points represent very low bandwidth storage nodes (~1Mbps), which likely do not download the entire file and DHP Early Exit does not occur.

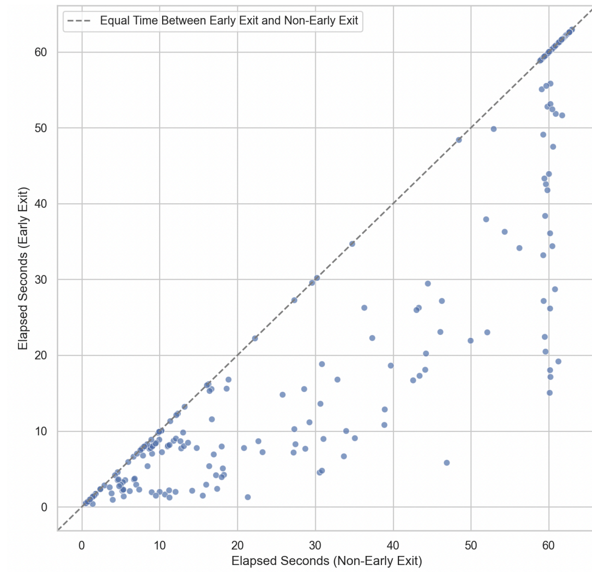

Figure 5: Scatter plot comparison between the elapsed seconds of DHP Early Exit v. non-Early Exit

Occasionally, the workers equally distribute the available bandwidth providing an accurate picture of the saturated bandwidth of the storage node regardless of the usage of DHP Early Exit.

In addition to the time savings, Figure 6 demonstrates the reduction in total volume of data downloaded comparing EE and non-EE downloads.

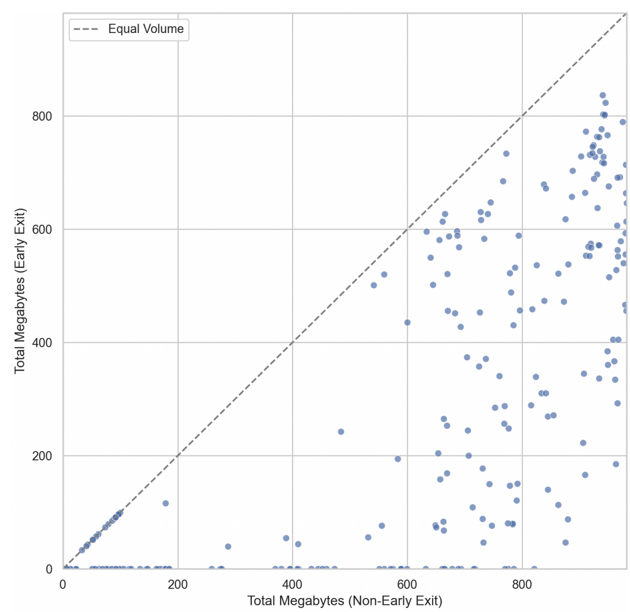

Figure 6: Scatter plot comparison between total volume downloaded of DHP Early Exit v. non-Early

The y-axis is the cumulative megabytes downloaded in a measurement session using DHP Early Exit, and the x-axis is the cumulative megabytes downloaded without

DHP EE. The dashed horizontal line represents points where the equal volume of data is downloaded. These values are generally circumstances where DHP EE is not applied.

Figure 6 only visualizes measurement sessions where at most 10 workers are used. This removes 7 outlier points provided by the AWS data centers. These values ranged from 2GB to 5GB of total volume downloaded as they had up to 40 workers. Only the value appearing within this graph will be considered for volume comparison analysis.

In figure 6, the 191 downloaded a total volume of **~151.72 gigabytes**. After simulating DHP Early Exit, the total volume downloaded dropped to **~96.88 gigabytes**, a **~54.84 gigabyte difference.** When considering the method of simulation strictly uses second by second logs, this volume save may be even larger.

4.4 Discussion

The results from the simulations demonstrate that the DHP Early Exit strategy maintains a relatively minimal difference in reported bandwidth when compared to the current method. DHP EE also drastically reduces elapsed time by nearly ⅙ **of the total duration** and reduces the total volume of data downloaded by **36.15%**. This section reflects on the practical advantages of Early Exit, evaluates its tradeoffs, and outlines considerations for its integration into Filecoin's bandwidth measurement system.

4.4.2 Practical Advantages of Early Exit

The Early Exit strategy addresses two primary shortcomings of the current BMS pipeline: bandwidth overestimation and unnecessarily long measurement durations. By terminating a measurement session once the first worker completes its download, DHP Early Exit reduces the likelihood that remaining workers consume reclaimed bandwidth, which otherwise inflates the reported measurement. This effect was consistently observed across simulations.

In simulations where bandwidth inflation was not a concern, DHP Early Exit provided a reported bandwidth within a reasonable value of the current method and a reduction in measurement duration and total volume downloaded. On average, **5.78 seconds** are saved and the total volume downloaded is reduced by **287.14 megabytes** per measurement.

Given that AWS Fargate ECS (and similar services) charge on a volume over time model, this reduction in total volume downloaded and elapsed time may lead to lower operational costs.

4.4.2 Limitations and Tradeoffs

Despite its benefits, DHP Early Exit introduces complexity into the BMS architecture. The current system does not support a mid-download termination of containerized worker processes, and modifying this behavior would require new coordination mechanisms among workers and job controllers. Such changes increase the risk of introduction of software bugs or incomplete measurements.

Moreover, DHP Early Exit is not universally applicable. In some measurements, no worker completes the download due to low bandwidth providers (*up to 109Mbps*). In these cases, Early Exit cannot be triggered and measurements revert to the original BMS behavior. Conversely, for high-bandwidth storage nodes, the system may require larger file sizes to ensure that multiple workers contribute meaningful throughput data before the earliest completion triggers an exit. Without this adjustment, bandwidth may be underestimated due to premature termination before worker performance stabilizes.

These tradeoffs suggest that a static file size or exit threshold may not be optimal across the full range of node performance. An adaptive strategy–modifying file size or exit criteria based on initial throughput estimates–may offer improved accuracy and robustness.

6. FURTHER EXPLORATION

With these findings, Filecoin's bandwidth measurement system intends to implement more elaborate services for the purpose of better coordinating workers. They also intend to implement the DHP Early Exit strategy into the BMS pipeline. For the next iteration of the system, substantially more data will be available resulting in further questions to be explored.

New available data will allow for the measurement of consistency in storage nodes. An important consideration is concerning the amount of measurements required to gain a high confidence level of the storage provider's available bandwidth. How often does a node need to be measured, and under what variance in conditions (i.e. different locations of workers, amount of workers) must it be measured to know that its bandwidth is saturated within a factor of two? Hopefully once this question is answered, we can reduce the load on the network from the repeated measurements and reduce the cost of measurement data in general.

In the updated system, a field containing the IP address of the storage node is expected to be added to the job data. Using this IP data, the impact of location on the available bandwidth of the storage node can be better understood. More specifically, we can hopefully visualize whether ISPs are a determinant of the expected performance of a storage node. By using the target URLs in the current data, some IP and ISP information is obtainable. Of this data set, we found a trend where the consistently lowest-performing nodes belonged to Alibaba. We characterized 3 storage nodes which were measured 32 times. Of these, the median value of their download speed is ~1.05Mbps.

A possible reason as to Alibaba's low performance could result from China's Great Firewall acting as a bottleneck. The BMS in its current form is not using workers from within China, instead they are located in Amazon data centers on the US east coast, in Hong Kong, and in Singapore. In the case of AWS, the incredibly high throughput seen in Figure 4 is a result of the workers measuring from within the same data center as the storage node. The workers are not measuring potential bottlenecks of the storage system, but instead the internal speed of AWS. Both the issues regarding Alibaba's performance and AWS's performance are potentially solved by increasing the diversity of locations from where BMS measures.

7. CONCLUSIONS

As Filecoin's bandwidth measurement system evolves, the conflict between cost and accuracy continues. The adoption of a DHP Early Exit strategy fulfills both of these considerations and will result in a necessary complexity overhead in the BMS pipeline. With better communication between workers, more complicated tasks which assist in deeper analyses of Filecoin's network of storage nodes. The findings in this paper led to a realization of inaccuracies and unnecessary work occurring due to BMS's current measurement pipeline.

8. ACKNOWLEDGEMENTS

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https://www.researchgate.net/publication/228944996_Throughput_and_fairness_of_multiple_TCP_connections_in_wireless_networks (TCP unfairness)